Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/26/2006
PI Name:	Overland, David M.S.		
Project Title:	Advanced Control System Development for Integration of Advanced Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	david.overland-1@nasa.gov	**Fax:**	FY 281-483-3204
PI Organization Type:	NASA CENTER	**Phone:**	281-483-4304
Organization Name:	NASA - Johnson Space Center		
PI Address 1:	Johnson Space Center		
PI Address 2:	Mail Code ER2		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/1998	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Edeen, Marybeth		
COI Name (Institution):	Hoo, Karlene A (Texas Tech University)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The purpose of this project is to develop advanced control system technology as part of an integrated approach to the command and control system of advanced life support. The control strategies for ALS systems must deal with continuous and discrete processes and with the dynamical interactions between sub-systems such as air revitalization, water recovery, food production, solids processing and the crew. The goal of autonomously controlling an ALS system challenges many areas of research, including software engineering, distributed data management and control, sensor interpretation, planning and scheduling, modeling and simulation, and validation and verification of autonomous control systems. These various research areas must be integrated into a coherent system that runs day after day for years and that can effectively interact with crewmembers, ground support and other systems. The control strategy must be able to reach “across” the system and “down” into its parts to gather information necessary to achieve mission objectives. Interfaces to crew, ground control, and other spacecraft systems must provide management of, and insight into		

Task Description:	operations, while still supporting human tasks. The current task objective is to determine automation and autonomy technology gaps and risks, and to ultimately devise a roadmap for developing required technology and mitigating those risks. Advanced control system technology is a necessary discipline for NASA exploration missions, yet because it is also a new discipline, not previously required for human spaceflight, technology risks and gaps are not brought forward in exploration mission planning. System Engineering and Integration (SE&I) of advanced controls is a critical technology gap identified by the Bioastronautic Critical Path Roadmap (Question 40a-i, and Questions 29 (bcehijk), 36 (cef), 38 (abcd), 39 (abc), 41 (ij), 42 (k), 43 (defgh), 44 (g), 47 (efghil), 48 (aefg), 49 (g), and 50 (bcdefgh). This technology gap crosses multiple disciplines. This current task is the only one directly addressing the identification of technology needs and capabilities required to close the gap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will benefit development of integrated control systems for large complex integrated systems where human intervention needs to be limited, but available on demand. Possible applications include remote processing plants or work in hostile environments. The technology also enables deeper integration of process goals and operations with continuous control and automation. This capability has great potential in the process control industry. A third benefit is the impact on software and system engineering processes as we learn to integrate large complex systems that also require complex advanced software.
Task Progress:	- The NASA Workshop on Advanced System Integration and Control for Life Support (ASICL) was held in August, 2003. The Exploration Systems Mission Division Life Support & Habitation Program issued the Workshop Final Report on January 18, 2005. - PI is continuing to actively participate as an Interdisciplinary Team member of the Advanced Integration Matrix (AIM) Program. Team Bravo was formed under AIM to investigate issues associated with advanced control and software systems integration. - The AIM Benchmark Test was renamed the AIM Controls Integration Test. One of the two bioreactors has been constructed, having passed Critical Design and Safety reviews. The second bioreactor necessitated a redesign as a result of CDR, postponing the integrated test until FY06. - The FY04 NASA Summer Faculty Fellowship professor, Dr. Karlene Hoo from Texas Tech, continued to support the process and controls design portion of the AIM Test. Additional model refinement and verification testing was conducted at TTU by a graduate student, Xi Zhang, in cooperation with the TTU civil engineering testbed efforts conducted under the ALS program. - The paper “Modeling and Control Studies of an Integrated Biological Wastewater Treatment System” was presented at the Society for Automotive Engineers (SAE) 35th International Conference for Environmental Systems (ICES) in Rome, Italy. This paper reported on the modeling and control study performed on Water Recovery System bioreactor prototypes. This study is still ongoing. - Small Business Innovative Research (SBIR) Subtopic B3.08 - Software Architectures and Integrated Control Strategies for Advanced Life Support Systems completed FY 05 Phase II and Phase I reviews. Twelve proposals were received for the Phase I; seven were recommended. No contracts were awarded. - SBIR B3.06-8618 “ISS Inventory Tracking System” Phase I completed. DO5 Cargo Integration and Operations Branch helped monitor the task and concurred with our recommendation for a Phase II. No contract was awarded. - With the restructuring of the SBIR program to align with Exploration goals, SBIR Subtopics X12.04 Advanced Environmental Monitoring and Control, X13.03 Software and Systems Engineering, and X14.09 Multi-use Microgravity and Software were developed and managed by the PI. “Process and Tool Support for Ontology-Aware Life Support System Development and Integration” Intelligent Automation, Inc. 05-1-X12.04-9861 JSC was awarded a Phase I contract. - A Center Director Discretionary Fund (CDDF) proposal, “Control System for Integration of Regenerative Life Support” was made to enhance the controls testbed development but was not funded.
Bibliography Type:	Description: (Last Updated: 06/19/2006)
Papers from Meeting Proceedings	Zhang X, Hoo KA, Overland DA. "Modeling and Control Studies of an Integrated Biological Wastewater Treatment System." SAE 35th International Conference on Environmental Systems (ICES) Conference, Rome Italy, July 2005. SAE Paper, July 2005. , Jul-2005